UNIVERSIDAD NACIONAL DE INGENIERÍA
Facultad de Ciencias
Escuela Profesional de Ciencia de la Computación

QUINTA PRÁCTICA CALIFICADA

Curso: **Fundamentos de Programación CC112 - A**

Nombres y Apellidos: Leo Alonso Velásquez Ronto

Código: 20252226H

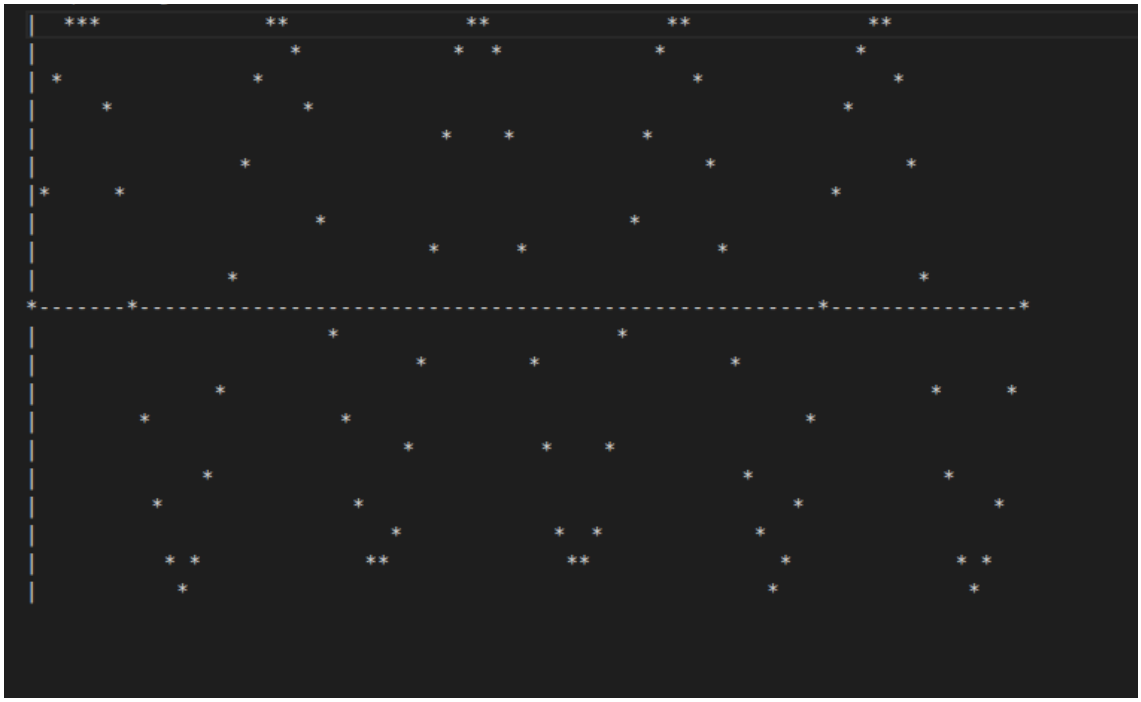
1.

```
pcs5@pcs5:~/Documentos/Fundamentos/pc02-cc112a-LeoV32/semana7/solpc5$ make
make: No se hace nada para 'all'.
pcs5@pcs5:~/Documentos/Fundamentos/pc02-cc112a-LeoV32/semana7/solpc5$ make asan
make: No se hace nada para 'asan'.
pcs5@pcs5:~/Documentos/Fundamentos/pc02-cc112a-LeoV32/semana7/solpc5$ make valgrind
valgrind --leak-check=full --show-leak-kinds=all ./build/debug/prob1
==10898== Memcheck, a memory error detector
==10898== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==10898== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==10898== Command: ./build/debug/prob1
==10898==
Se genero el archivo correctamente
==10898==
==10898== HEAP SUMMARY:
==10898==     in use at exit: 0 bytes in 0 blocks
==10898==   total heap usage: 97 allocs, 97 frees, 97,242 bytes allocated
==10898==
==10898== All heap blocks were freed -- no leaks are possible
==10898==
==10898== For lists of detected and suppressed errors, rerun with: -s
==10898== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
pcs5@pcs5:~/Documentos/Fundamentos/pc02-cc112a-LeoV32/semana7/solpc5$ ./build/debug/prob1
Se genero el archivo correctamente
pcs5@pcs5:~/Documentos/Fundamentos/pc02-cc112a-LeoV32/semana7/solpc5$
```

a7 > solpc5 > reporteAlumnos.txt

```
1001    Juan        15.6667     APROBADO
1002    Maria       10.6667     APROBADO
1003    Luis        19      APROBADO
1004    Ana     8.33333     DESAPROBADO
1005    Carlos      15.3333     APROBADO
1006    Lucia       12      APROBADO
1007    Pedro       6       DESAPROBADO
1008    Sofia       19      APROBADO
1009    Diego       15      APROBADO
1010    Valeria     9       DESAPROBADO
1011    Andres      13      APROBADO
1012    Carmen      17      APROBADO
1013    Ricardo     9       DESAPROBADO
1014    Elena       19      APROBADO
1015    Miguel      6       DESAPROBADO
1016    Patricia        15      APROBADO
1017    Fernando        11      APROBADO
1018    Daniela     19.6667     APROBADO
1019    Jorge       5       DESAPROBADO POR REGLA ACADEMICA
1020    Paola       14      APROBADO
1021    Marco       12      APROBADO
1022    Rosa        8       DESAPROBADO
1023    Julio       17      APROBADO
1024    Teresa      10      APROBADO
1025    Hugo        18      APROBADO
1026    Alicia      13      APROBADO
1027    Bruno       6       DESAPROBADO
1028    Claudia     15      APROBADO
1029    Ivan        9       DESAPROBADO
1030    Monica      17      APROBADO

Total de estudiantes: 30
Promedio general: 12.7889
Estudiante con mayor promedio: Daniela  19.6667
Estudiante con menor promedio: Jorge    5
```

3.

```
root@DESKTOP-OQQMUKA:~/proyectos/pc02-cc112a-LeoV32/semana7/pc5cc112# make
mkdir -p build/normal
g++ -std=c++17 -Wall -Wextra -Wpedantic prob3.cpp -o build/normal/prob3
root@DESKTOP-OQQMUKA:~/proyectos/pc02-cc112a-LeoV32/semana7/pc5cc112# make asan
mkdir -p build/asan
g++ -std=c++17 -Wall -Wextra -Wpedantic -fsanitize=address -fno-omit-frame-pointer
-g -O1 prob3.cpp -o build/asan/prob3
root@DESKTOP-OQQMUKA:~/proyectos/pc02-cc112a-LeoV32/semana7/pc5cc112# make valgrind

mkdir -p build/debug
g++ -std=c++17 -Wall -Wextra -Wpedantic -g -O0 prob3.cpp -o build/debug/prob3
valgrind --leak-check=full --show-leak-kinds=all ./build/debug/prob1
==10348== Memcheck, a memory error detector
==10348== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==10348== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==10348== Command: ./build/debug/prob1
==10348==
Se genero el archivo correctamente
==10348==
==10348== HEAP SUMMARY:
==10348==     in use at exit: 0 bytes in 0 blocks
==10348==   total heap usage: 97 allocs, 97 frees, 97,242 bytes allocated
==10348==
==10348== All heap blocks were freed -- no leaks are possible
==10348==
==10348== For lists of detected and suppressed errors, rerun with: -s
==10348== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@DESKTOP-OQQMUKA:~/proyectos/pc02-cc112a-LeoV32/semana7/pc5cc112# ./build/debug
/prob3
Se genero el archivo grafico.txt
root@DESKTOP-OQQMUKA:~/proyectos/pc02-cc112a-LeoV32/semana7/pc5cc112#
```